**RAFFLES INSTITUTION**
**H2 Mathematics 9758**
**2026 Year 6 Term 3 Revision 5 (Lambda)**
**Topic(s): Vectors**
**(Vectors II and III (Solutions))**

**Source of Question: JPJC Prelim 9758/2022/02/Q2**

**1** **(a)** With reference to the origin $O$, the points $A$, $B$ and $X$ are $\overrightarrow{OA}=\mathbf{a}$ , $\overrightarrow{OB}=\mathbf{b}$ and $\overrightarrow{OX}=\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}$. The point $Y$ lies on $AB$ such that $O$, $X$ and $Y$ are collinear. Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio of $AY:YB$. [5]

**(b)** The points $P$, $Q$ and $R$ have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively. $P$ and $Q$ are fixed, and $R$ varies. Describe geometrically the set of possible positions of the point $R$ such that

**(i)** $(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$, [2]

**(ii)** $(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$. [2]

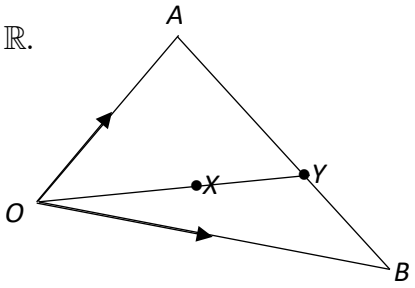
**Solution:**

**1(a)**

$Y$ lies on line $AB \Rightarrow \overrightarrow{OY}=\mathbf{a}+\beta(\mathbf{b}-\mathbf{a})$, for some $\beta\in\mathbb{R}$.

Since $O$, $X$ and $Y$ are collinear,

$\mathbf{a}+\beta(\mathbf{b}-\mathbf{a})=\alpha\left(\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}\right)$ for some $\alpha\in\mathbb{R}$.




Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel,

Comparing,

$1-\beta=\frac{\alpha}{8}$ ---(1)

$\beta=\frac{3\alpha}{8}$ ---(2)

Solving (1) and (2), $\alpha=2$, $\beta=\frac{3}{4}$

$\overrightarrow{OY}=2\left(\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}\right)=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$

$\overrightarrow{AY}=\frac{3}{4}\overrightarrow{AB}$

$AY:YB=3:1$

| | |
|---|---|
| **(b)(i)** | $(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$<br>We see that either $(\mathbf{r}-\mathbf{p})//\mathbf{q}$ or $(\mathbf{r}-\mathbf{p})=\mathbf{0}$<br>$\mathbf{r}=\mathbf{p}+k\mathbf{q},\ k\neq 0$ or $\mathbf{r}=\mathbf{p}$<br>Thus $\mathbf{r}=\mathbf{p}+k\mathbf{q},\ k\in\mathbb{R}$<br>The set of all possible positions of the point $R$ is the line that passes through the point $P$ and parallel to the vector $\mathbf{q}$. |
| **(b)(ii)** | $(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$<br>$\mathbf{r}.\mathbf{q}-\mathbf{p}.\mathbf{q}=0$<br>$\mathbf{r}.\mathbf{q}=\mathbf{p}.\mathbf{q}$<br>The set of all possible positions of the point $R$ is the plane that is perpendicular to $\mathbf{q}$ and containing the point $P$. |

**Source of Question: EJC Prelim 9758/2025/P1/Q6**

**2** There are three distinct points $A$, $B$ and $P$. The point $P$ lies on the circle with centre $O$ and diameter $AB$. Relative to the origin $O$, the position vectors of points $A$ and $P$ are **a** and **p** respectively.

**(a)** By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of **a** and **p**, show that $\overrightarrow{AP}.\overrightarrow{BP}=0$. [3]

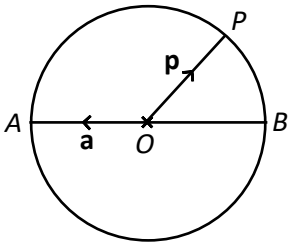
The point $C$ with position vector **c** lies on $AB$ produced such that $AC:AB$ is $\lambda:1$.

**(b)** Find **c** in terms of $\lambda$ and **a**. [2]

**(c)** It is given that $PC$ is a tangent to the circle and that angle $AOP$ is $120°$. Using a suitable scalar product, find the value of $\lambda$. [4]

**Solution:**

**(a)**




$\overrightarrow{AP}=\mathbf{p}-\mathbf{a}$

$\overrightarrow{BP}=\mathbf{p}+\mathbf{a}$

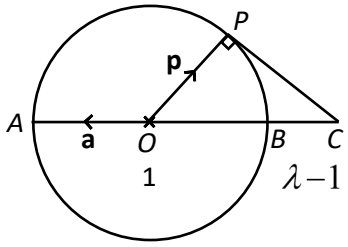
$\overrightarrow{AP}.\overrightarrow{BP}=(\mathbf{p}-\mathbf{a}).(\mathbf{p}+\mathbf{a})$

$=|\mathbf{p}|^2-|\mathbf{a}|^2$

$=|\mathbf{a}|^2-|\mathbf{a}|^2$ (since $OA$ and $OP$ are radius, $\therefore|\mathbf{p}|=|\mathbf{a}|$)

$=0$

**(b)**




By Ratio Theorem,

$\mathbf{b}=\dfrac{(\lambda-1)\mathbf{a}+\mathbf{c}}{\lambda}$

$\mathbf{c}=\lambda\mathbf{b}-(\lambda-1)\mathbf{a}$

$\mathbf{c}=\lambda(-\mathbf{a})-(\lambda-1)\mathbf{a}$

$=(1-2\lambda)\mathbf{a}$

| | |
|---|---|
| **(c)** | $\overrightarrow{PC} = (1-2\lambda)\mathbf{a} - \mathbf{p}$<br><br>$\overrightarrow{OP}.\overrightarrow{PC} = 0$<br>$\mathbf{p}.((1-2\lambda)\mathbf{a} - \mathbf{p}) = 0$<br>$(1-2\lambda)\mathbf{p}.\mathbf{a} - \mathbf{p}.\mathbf{p} = 0$<br>$(1-2\lambda)\lvert\mathbf{p}\rvert\lvert\mathbf{a}\rvert\cos 120^\circ - \lvert\mathbf{p}\rvert^2 = 0$<br>$-\frac{1}{2}(1-2\lambda)\lvert\mathbf{a}\rvert^2 - \lvert\mathbf{a}\rvert^2 = 0$<br>$-\frac{1}{2}(1-2\lambda) = 1$<br>$\lambda = \frac{3}{2}$ |

**Source of Question: NJC Prelim 9758/2024/01/Q4**

**3** Relative to the origin $O$, points $A$, $B$ and $C$ have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-zero vectors that are not parallel to one another. The points $A$, $B$ and $C$ are not collinear.

A point of trisection is a point that divides a line segment internally in the ratio $1:2$ or $2:1$. Suppose another two points $D$ and $E$ are points of trisection of line segments $AB$ and $AC$ respectively and both points are nearer to $A$ than to $B$ and $C$ respectively. The lines $BE$ and $CD$ meet at point $F$.

**(i)** Show that the vector equations of the lines $BE$ and $CD$ can be expressed as $\mathbf{r}=\frac{2}{3}\lambda\mathbf{a}+(1-\lambda)\mathbf{b}+\frac{1}{3}\lambda\mathbf{c}$ and $\mathbf{r}=\frac{2}{3}\mu\mathbf{a}+\frac{1}{3}\mu\mathbf{b}+(1-\mu)\mathbf{c}$ respectively, where $\lambda$ and $\mu$ are parameters. Hence, show that at point $F$, $\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}=\mathbf{0}$, where $\alpha$, $\beta$ and $\gamma$ are constants, each to be expressed in terms of $\lambda$ and $\mu$. [4]

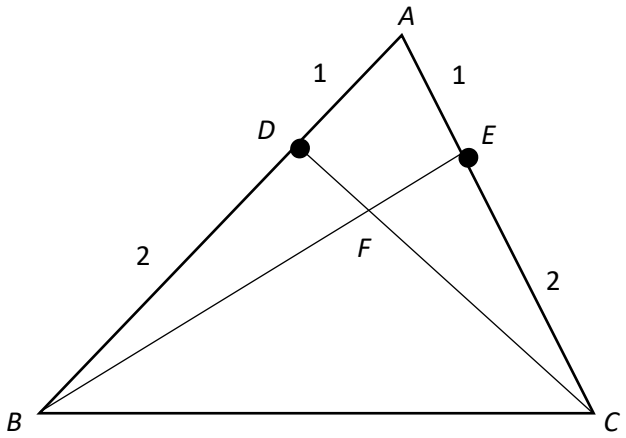
**(ii)** Given further that $OACB$ is a parallelogram, find the position vector of $F$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [3]

**Solution:**

**(i)**

Using Ratio Theorem,

$$\overrightarrow{BE}$$
$$=\frac{1}{3}\left(\overrightarrow{BC}+2\overrightarrow{BA}\right)$$
$$=\frac{1}{3}\left(\mathbf{c}-\mathbf{b}+2\left(\mathbf{a}-\mathbf{b}\right)\right)$$
$$=\frac{1}{3}\mathbf{c}+\frac{2}{3}\mathbf{a}-\mathbf{b}$$
$$\overrightarrow{CD}$$
$$=\frac{1}{3}\left(\overrightarrow{CB}+2\overrightarrow{CA}\right)$$
$$=\frac{1}{3}\left(\mathbf{b}-\mathbf{c}+2\left(\mathbf{a}-\mathbf{c}\right)\right)$$
$$=\frac{1}{3}\mathbf{b}+\frac{2}{3}\mathbf{a}-\mathbf{c}$$



| | |
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| | $l_{BE}: \mathbf{r} = \mathbf{b} + \lambda\left(\frac{1}{3}\mathbf{c} + \frac{2}{3}\mathbf{a} - \mathbf{b}\right)$<br>$= \frac{2}{3}\lambda\mathbf{a} + (1-\lambda)\mathbf{b} + \frac{1}{3}\lambda\mathbf{c}, \lambda \in \mathbb{R}$ (shown)<br>$l_{CD}: \mathbf{r} = \mathbf{c} + \mu\left(\frac{1}{3}\mathbf{b} + \frac{2}{3}\mathbf{a} - \mathbf{c}\right)$<br>$= \frac{2}{3}\mu\mathbf{a} + \frac{1}{3}\mu\mathbf{b} + (1-\lambda)\mathbf{c}$, $\mu \in \mathbb{R}$ (shown)<br>At $F$, $\frac{2}{3}\lambda\mathbf{a} + (1-\lambda)\mathbf{b} + \frac{1}{3}\lambda\mathbf{c} = \frac{2}{3}\mu\mathbf{a} + \frac{1}{3}\mu\mathbf{b} + (1-\lambda)\mathbf{c}$<br>$\left(\frac{2}{3}\lambda - \frac{2}{3}\mu\right)\mathbf{a} + \left(1 - \lambda - \frac{1}{3}\mu\right)\mathbf{b} + \left(\frac{1}{3}\lambda + \mu - 1\right)\mathbf{c} = \mathbf{0}$ (shown)<br>where $\alpha = \frac{2}{3}\lambda - \frac{2}{3}\mu, \quad \beta = 1 - \lambda - \frac{1}{3}\mu, \quad \gamma = \frac{1}{3}\lambda + \mu - 1$ |
| **(ii)** | If $OACB$ is a parallelogram,<br>$\mathbf{c} = \mathbf{a} + \mathbf{b}$<br>$\left(\frac{2}{3}\lambda - \frac{2}{3}\mu\right)\mathbf{a} + \left(1 - \lambda - \frac{1}{3}\mu\right)\mathbf{b} + \left(\frac{1}{3}\lambda + \mu - 1\right)\mathbf{c} = \mathbf{0}$<br>$\left(\frac{2}{3}\lambda - \frac{2}{3}\mu\right)\mathbf{a} + \left(1 - \lambda - \frac{1}{3}\mu\right)\mathbf{b} + \left(\frac{1}{3}\lambda + \mu - 1\right)(\mathbf{a} + \mathbf{b}) = \mathbf{0}$<br>$\left(\frac{2}{3}\lambda - \frac{2}{3}\mu + \frac{1}{3}\lambda + \mu - 1\right)\mathbf{a} + \left(1 - \lambda - \frac{1}{3}\mu + \frac{1}{3}\lambda + \mu - 1\right)\mathbf{b} = \mathbf{0}$<br>$\left(\lambda + \frac{1}{3}\mu - 1\right)\mathbf{a} + \left(\frac{2}{3}\mu - \frac{2}{3}\lambda\right)\mathbf{b} = \mathbf{0}$<br><br>Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel to each other and are non-zero vectors,<br>$\lambda + \frac{1}{3}\mu - 1 = 0 \quad \cdots(1)$<br>$\frac{2}{3}\mu - \frac{2}{3}\lambda = 0 \quad \cdots(2)$<br>Solving, $\lambda = \mu = \frac{3}{4}$<br>Thus position vector of $F$ is given by<br>$= \mathbf{b} + \frac{3}{4}\left(\frac{1}{3}(\mathbf{a} + \mathbf{b}) + \frac{2}{3}\mathbf{a} - \mathbf{b}\right)$<br>$= \frac{3}{4}\mathbf{a} + \frac{1}{2}\mathbf{b}$ |

**Source of Question: EJC Prelim 9758/2022/01/Q7**

**4** The line $l_1$ has equation $\mathbf{r} = 11\mathbf{i} - 3\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$, where $\lambda$ is a parameter. The point $A$ has coordinates $(2, 0, -1)$.

**(a)** The plane $p$ contains the line $l_1$ and the point $A$. Find a cartesian equation of the plane $p$.

[3]

**(b)** Find the position vector of the point $A'$, the reflection of the point $A$ in the line $l_1$.

[4]

The line $l_2$ passes through the point $A$, and is perpendicular to the line $l_1$. Planes $p$ and $q$ are perpendicular planes such that $p$ and $q$ meet at line $l_2$.

**(c)** Find a vector equation of the line $l_2$.

[2]

**(d)** The plane $\Pi$ is such that $\Pi$ contains the point $(11, -3, -1)$ and is parallel to $q$. Find the perpendicular distance between $\Pi$ and $q$.

[2]

**Solution:**

**(a)**

Two vectors parallel to $p$ are $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 11 \\ -3 \\ -1 \end{pmatrix} = 3\begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}$.

$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ -2 \end{pmatrix} = -2\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

Thus a vector normal to $p$ is $\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

Since $A(2, 0, -1)$ lies on $p$, equation of $p$ is $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = 1$,

a cartesian equation of $p$ is $x + 3y + z = 1$.

**(b)**

Let $G$ be the foot of perpendicular from $A$ to $l_1$.

Since $G$ lies on $l_1$, $\overrightarrow{OG} = \begin{pmatrix} 11 \\ -3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$.

$$\overrightarrow{AG}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=0\Rightarrow\left[\begin{pmatrix}11\\-3\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\-1\\2\end{pmatrix}-\begin{pmatrix}2\\0\\-1\end{pmatrix}\right]\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=0$$

$$\Rightarrow 12+6\lambda-0=0$$

$$\Rightarrow \lambda=-2$$

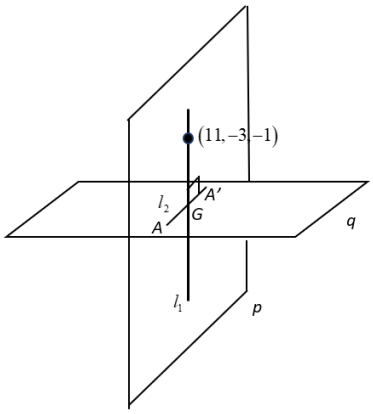
Hence, $\overrightarrow{OG}=\begin{pmatrix}9\\-1\\-5\end{pmatrix}$.

Let $A'$ be the point of reflection of $A$ in line $l_1$.

Using Ratio Theorem, $\overrightarrow{OG}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$.

$$\therefore\quad \overrightarrow{OA'}=2\overrightarrow{OG}-\overrightarrow{OA}=2\begin{pmatrix}9\\-1\\-5\end{pmatrix}-\begin{pmatrix}2\\0\\-1\end{pmatrix}=\begin{pmatrix}16\\-2\\-9\end{pmatrix}.$$

**(c)**




A direction vector of $l_2$ is $\overrightarrow{AG}=\begin{pmatrix}7\\-1\\-4\end{pmatrix}$.

A vector equation of $l_2$ is

$$\mathbf{r}=\begin{pmatrix}2\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}7\\-1\\-4\end{pmatrix},\quad \mu\in\mathbb{R}.$$

**Alternative : $l_2$ is the line of intersection between $p$ and $q$**

Equation of $p$: $x+3y+z=1$ --- (1)

Equation of $q$: $\mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=\begin{pmatrix}2\\0\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=0\Rightarrow x-y+2z=0$ --- (2)

Using GC to solve (1) and (2), an equation of $l_2$ is $\mathbf{r}=\begin{pmatrix}\frac{1}{4}\\\frac{1}{4}\\0\end{pmatrix}+\mu\begin{pmatrix}-\frac{7}{4}\\\frac{1}{4}\\1\end{pmatrix},\quad \mu\in\mathbb{R}$.

**(d)**

Note that the points $G(9,-1,-5)$ on $q$ and $B(11,-3,-1)$ on $\prod$, lie on the same line $l_1$, and $l_1$ is perpendicular to both $q$ and $\prod$.

Perpendicular distance between $\prod$ and $q$ $=\left|\overrightarrow{GB}\right|=\left|\begin{pmatrix}11\\-3\\-1\end{pmatrix}-\begin{pmatrix}9\\-1\\-5\end{pmatrix}\right|=\left|\begin{pmatrix}2\\-2\\4\end{pmatrix}\right|=2\sqrt{6}$ units

**Alternative 1 : Use distance between parallel planes formula**

Plane $\prod$ : $\mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=\begin{pmatrix}11\\-3\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=12$ Plane $q$: $\mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=0$

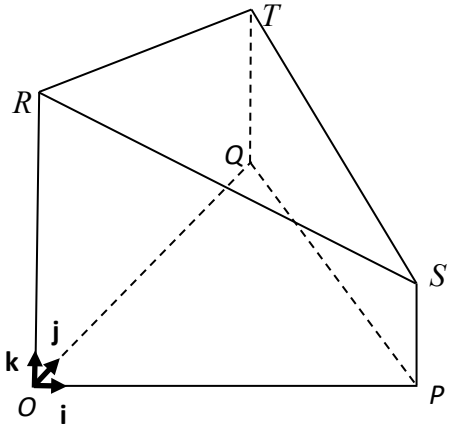
Perpendicular distance $=\dfrac{|d_1-d_2|}{|\mathbf{n}|}=\dfrac{|12-0|}{\left|\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}=\dfrac{12}{\sqrt{6}}=2\sqrt{6}$ units

**Alternative 2 : Find projection length of $\overrightarrow{OB}$ on normal since $q$ contains $O$)**

Perpendicular distance $=\dfrac{\left|\overrightarrow{OB}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}=\dfrac{\left|\begin{pmatrix}11\\-3\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}{\left|\sqrt{6}\right|}=\dfrac{12}{\sqrt{6}}=2\sqrt{6}$ units

**Source of Question: ACJC CT2 9758/2023/01/Q8**

**5**




A tent is constructed with three vertical poles situated at $O$, $P$ and $Q$ on the horizontal ground. The poles $OR$, $PS$ and $QT$ have heights 16m, 6m and 8m respectively. It is given that $OP$ = 15m and $OQ$ = 20m. The point $O$ is taken as the origin, with unit vectors **i** along $OP$, **j** along $OQ$ and **k** along $OR$.

**(i)** Find, in the form $ax + by + cz = 240$, the equation of the plane $RST$. [3]

**(ii)** Find the angle between the plane $RST$ and the horizontal ground. [2]

**(iii)** Find the perpendicular distance from $P$ to the plane $RST$. [2]

**(iv)** To stabilise the tent, one end of a bar of fixed length 20m is attached to the point $R$, with the other end of the bar to be attached to a point $A$ on $PQ$. Find the coordinates of $A$. [4]

**Solution:**

**(i)**

$$\overrightarrow{OR} = \begin{pmatrix} 0 \\ 0 \\ 16 \end{pmatrix}, \quad \overrightarrow{OS} = \begin{pmatrix} 15 \\ 0 \\ 6 \end{pmatrix}, \quad \overrightarrow{OT} = \begin{pmatrix} 0 \\ 20 \\ 8 \end{pmatrix}$$

$$\overrightarrow{RS} \times \overrightarrow{RT} = \begin{pmatrix} 15 \\ 0 \\ -10 \end{pmatrix} \times \begin{pmatrix} 0 \\ 20 \\ -8 \end{pmatrix} = 5\begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \times 4\begin{pmatrix} 0 \\ 5 \\ -2 \end{pmatrix} = 20\begin{pmatrix} 10 \\ 6 \\ 15 \end{pmatrix}$$

Equation of plane $RST$:

$$\mathbf{r} \cdot \begin{pmatrix} 10 \\ 6 \\ 15 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 16 \end{pmatrix} \cdot \begin{pmatrix} 10 \\ 6 \\ 15 \end{pmatrix} \Rightarrow \mathbf{r} \cdot \begin{pmatrix} 10 \\ 6 \\ 15 \end{pmatrix} = 240$$

$\therefore\ 10x + 6y + 15z = 240$ where $a = 10$, $b = 6$, $c = 15$

**(ii)** Let $\theta$ be the angle between plane $RST$ and the horizontal ground.

| | |
|---|---|
| | $\cos\theta = \dfrac{\begin{pmatrix}10\\6\\15\end{pmatrix}}{\sqrt{361}} \cdot \dfrac{\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{1}} = \dfrac{15}{19}$<br>$\theta = 37.9^\circ$ (to 1 d.p.) |
| **(iii)** | Perpendicular distance from $P$ to plane $RST = \left\lvert \overrightarrow{PS} \cdot \hat{\mathbf{n}} \right\rvert$<br>$= \left\lvert \begin{pmatrix}0\\0\\6\end{pmatrix} \cdot \dfrac{1}{\sqrt{361}} \begin{pmatrix}10\\6\\15\end{pmatrix} \right\rvert$<br>$= \dfrac{90}{19}$ m $= 4.74$ m (3sf)<br><u>OR (longer method)</u><br>Let $F$ be the foot of the perpendicular from $P$ to plane $RST$.<br>Vector equation of line $PF$: $\mathbf{r} = \begin{pmatrix}15\\0\\0\end{pmatrix} + \lambda \begin{pmatrix}10\\6\\15\end{pmatrix}, \lambda \in \mathbb{R}$<br>$\therefore \begin{pmatrix}15+10\lambda\\6\lambda\\15\lambda\end{pmatrix} \cdot \begin{pmatrix}10\\6\\15\end{pmatrix} = 240 \Rightarrow 150 + 100\lambda + 36\lambda + 225\lambda = 240$<br>$\Rightarrow \lambda = \dfrac{90}{361}$<br>$\overrightarrow{PF} = \dfrac{90}{361}\begin{pmatrix}10\\6\\15\end{pmatrix}$ $\left\lvert \overrightarrow{PF} \right\rvert = \dfrac{90}{361}\sqrt{10^2+6^2+15^2} = \dfrac{90}{19}$ m $= 4.74$ m (3sf) |
| **(iv)** | $\overrightarrow{PQ} = \begin{pmatrix}0\\20\\0\end{pmatrix} - \begin{pmatrix}15\\0\\0\end{pmatrix} = \begin{pmatrix}-15\\20\\0\end{pmatrix} = 5\begin{pmatrix}-3\\4\\0\end{pmatrix}$<br>Vector equation of line $PQ$: $\mathbf{r} = \begin{pmatrix}15\\0\\0\end{pmatrix} + \lambda \begin{pmatrix}-3\\4\\0\end{pmatrix}, \lambda \in \mathbb{R}$<br>Since $A$ lies on $PQ$, $\overrightarrow{OA} = \begin{pmatrix}15\\0\\0\end{pmatrix} + \lambda \begin{pmatrix}-3\\4\\0\end{pmatrix}$, for some $\lambda \in \mathbb{R}$<br>$\overrightarrow{RA} = \begin{pmatrix}15-3\lambda\\4\lambda\\0\end{pmatrix} - \begin{pmatrix}0\\0\\16\end{pmatrix} = \begin{pmatrix}15-3\lambda\\4\lambda\\-16\end{pmatrix}$ |

$$\left|\overrightarrow{RA}\right| = 20 \Rightarrow \sqrt{(15-3\lambda)^2 + (4\lambda)^2 + 16^2} = 20$$

$$225 - 90\lambda + 9\lambda^2 + 16\lambda^2 + 256 = 400$$

$$25\lambda^2 - 90\lambda + 81 = 0 \Rightarrow (5\lambda - 9)^2 = 0$$

$$\lambda = \frac{9}{5}$$

$$\overrightarrow{OA} = \begin{pmatrix} 15 \\ 0 \\ 0 \end{pmatrix} + \frac{9}{5}\begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 48 \\ 36 \\ 0 \end{pmatrix} \quad \therefore A\left(\frac{48}{5}, \frac{36}{5}, 0\right)$$

**Source of Question: TMJC Prelim 9758/2025/P1/Q10**

**6** A drone flies in a straight line towards a rooftop for inspection. The drone's flight path is modelled by the line $l: \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$ where $\lambda \in \mathbb{R}$, and the rooftop is modelled by the plane $p: 2x + y - z = 7$.

**(a)** On Day 1, the drone flies towards the rooftop using the flight path. Upon reaching the rooftop, the drone flies in a new flight path, *m*. The acute angle between the original flight path and the normal to the rooftop is the same as the acute angle between the new flight path and the same normal to the rooftop. Both flight paths and the normal lie on the same plane. Find the cosine of the angle between the original flight path and the new flight path in an exact non-trigonometrical form. [3]

**(b)** On Day 2, the drone starts to descend from point *A* with coordinates $(2,1,0)$ towards the rooftop using the shortest possible path. Find the exact coordinates of the point where the drone lands on the rooftop. [4]

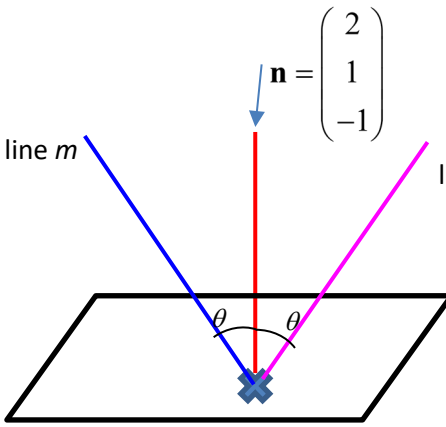
**(c)** After landing, the drone glides on the rooftop in another new flight path, *s*, that is perpendicular to the flight path *l*. Find a cartesian equation of the flight path *s*. [3]

A second rooftop is built parallel to the first rooftop *p* such that the perpendicular distance between these two rooftops is *d* units. Find two possible cartesian equations of the second rooftop in terms of *d*. [3]

**Solution**

**(a)**

Let $\theta$ be the acute angle between the direction vector of the flight path and normal vector of the rooftop.




$$\text{line } l: \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$$

$$p: \mathbf{r} \bullet \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = 7$$

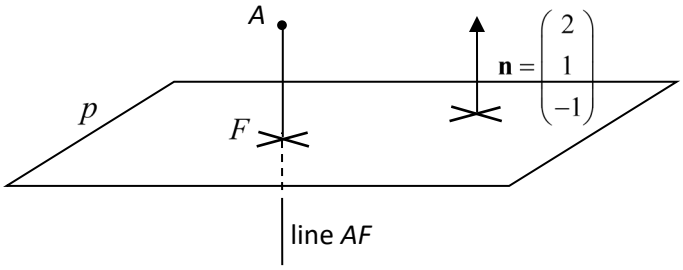
$$\cos\theta = \frac{\left| \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \right|}{\sqrt{14}\sqrt{6}} = \frac{3}{\sqrt{14}\sqrt{6}}$$

The angle between the original flight path and the new flight path will be $2\theta$.

$$\begin{aligned} \cos 2\theta &= 2\cos^2\theta - 1 \\ &= 2\left(\frac{3}{\sqrt{14}\sqrt{6}}\right)^2 - 1 \\ &= -\frac{11}{14} \end{aligned}$$

**(b)**

Let the foot of perpendicular be $F$.




$$l_{AF}: \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \ \mu \in \mathbb{R}$$

$$\begin{pmatrix}2+2\mu\\1+\mu\\-\mu\end{pmatrix}\bullet\begin{pmatrix}2\\1\\-1\end{pmatrix}=7$$

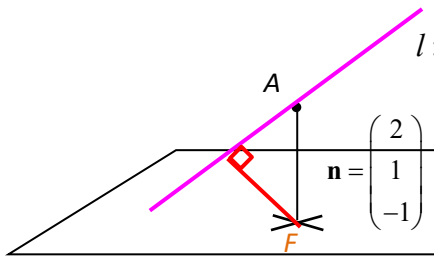
$$4+4\mu+1+\mu+\mu=7$$

$$6\mu+5=7$$

$$\mu=\frac{1}{3}$$

Coordinates of $F$ is $\left(\frac{8}{3},\frac{4}{3},-\frac{1}{3}\right)$.

**(c)**




$$l:\mathbf{r}=\begin{pmatrix}2\\1\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\-2\\1\end{pmatrix},\lambda\in\mathbb{R}$$

$$p:\mathbf{r}\bullet\begin{pmatrix}2\\1\\-1\end{pmatrix}=7$$

$$\mathbf{d}=\begin{pmatrix}3\\-2\\1\end{pmatrix}\times\begin{pmatrix}2\\1\\-1\end{pmatrix}=\begin{pmatrix}1\\5\\7\end{pmatrix}$$

$$l_s:\mathbf{r}=\begin{pmatrix}\frac{8}{3}\\\frac{4}{3}\\-\frac{1}{3}\end{pmatrix}+\alpha\begin{pmatrix}1\\5\\7\end{pmatrix},\ \alpha\in\mathbb{R}$$

$$x=\frac{8}{3}+\alpha\Rightarrow\alpha=\frac{3x-8}{3}$$

$$y=\frac{4}{3}+5\alpha\Rightarrow\alpha=\frac{3y-4}{15}$$

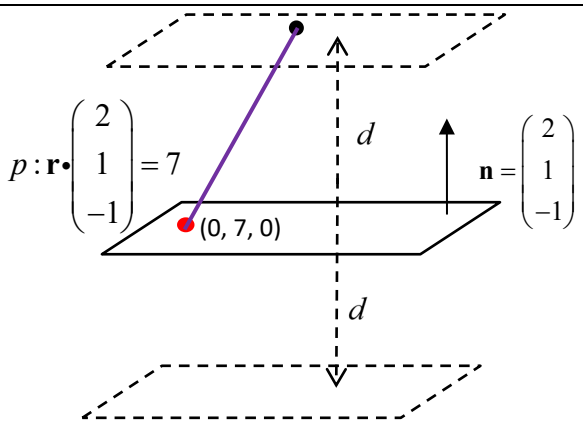
$$z=-\frac{1}{3}+7\alpha\Rightarrow\alpha=\frac{3z+1}{21}$$

$$\therefore\frac{3x-8}{3}=\frac{3y-4}{15}=\frac{3z+1}{21}$$

Let $(0,7,0)$ be a point on the first rooftop and $(x,y,z)$ be a point on the second rooftop.

Distance between 2 parallel rooftops: $(x,y,z)$

$$\frac{\left|\left[\begin{pmatrix}x\\y\\z\end{pmatrix}-\begin{pmatrix}0\\7\\0\end{pmatrix}\right]\bullet\begin{pmatrix}2\\1\\-1\end{pmatrix}\right|}{\sqrt{6}}=d$$




$$\frac{\left|\begin{pmatrix}x\\y-7\\z\end{pmatrix}\bullet\begin{pmatrix}2\\1\\-1\end{pmatrix}\right|}{\sqrt{6}}=d$$

$$|2x+y-7-z|=\sqrt{6}d$$

$2x+y-7-z=\sqrt{6}d$ or $2x+y-7-z=-\sqrt{6}d$

$2x+y-z=\sqrt{6}d+7$ or $2x+y-z=-\sqrt{6}d+7$

**Alternatively,**

$2x+y-z=7$

$$\mathbf{r}\cdot\begin{pmatrix}2\\1\\-1\end{pmatrix}=7$$

$$\mathbf{r}\cdot\frac{\begin{pmatrix}2\\1\\-1\end{pmatrix}}{\sqrt{6}}=\frac{7}{\sqrt{6}}$$

Let the second rooftop be:

$$\mathbf{r}\cdot\begin{pmatrix}2\\1\\-1\end{pmatrix}=D$$

$$\mathbf{r}\cdot\frac{\begin{pmatrix}2\\1\\-1\end{pmatrix}}{\sqrt{6}}=\frac{D}{\sqrt{6}}$$

Distance between 2 parallel rooftops:

$$\left|\frac{7}{\sqrt{6}}-\frac{D}{\sqrt{6}}\right|=d$$

$\frac{7}{\sqrt{6}}-\frac{D}{\sqrt{6}}=d$ or $-\frac{7}{\sqrt{6}}+\frac{D}{\sqrt{6}}=d$

$D=7-\sqrt{6}d$ or $D=7+\sqrt{6}d$

| | |
|---|---|
| | $\mathbf{r}\cdot\begin{pmatrix}2\\1\\-1\end{pmatrix}=\sqrt{6}d+7$ or $\mathbf{r}\cdot\begin{pmatrix}2\\1\\-1\end{pmatrix}=-\sqrt{6}d+7$<br>$2x+y-z=\sqrt{6}d+7$ or $2x+y-z=-\sqrt{6}d+7$ |

**Source of Question: ASRJC Prelim 9758/2025/P1/Q9**

**7** The line $\ell$ has the equation $\frac{x-a}{2} = y-2 = \frac{1-2z}{b}$, where $a$ and $b$ are real constants. The plane $\pi$ has the equation $3x-y+4z=10$.

**(a)** Given that the line $\ell$ and the plane $\pi$ do not intersect, show that $a \neq \frac{10}{3}$ and $b = \frac{5}{2}$. [5]

**(b)** It is given that $a=1$ and $b=3$. Find the point of intersection between the line $\ell$ and the plane $\pi$. [3]

**(c)** It is given now that $a=1$, $b=\frac{5}{2}$.

**(i)** Find the distance between the line $\ell$ and the plane $\pi$. [2]

**(ii)** Determine whether the line $\ell$ and the origin $O$ lie on the same side of the plane $\pi$. State an equation of the other plane that is equidistant from line $\ell$ and parallel to plane $\pi$. [3]

**Solution:**

**(a)**

$\frac{x-a}{2} = \frac{y-2}{1} = \frac{1-2z}{b} \quad \Rightarrow \quad \frac{x-a}{2} = \frac{y-2}{1} = \frac{z-\frac{1}{2}}{-\frac{b}{2}}$

Vector equation of line $\ell$: $\mathbf{r} = \begin{pmatrix} a \\ 2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -\frac{b}{2} \end{pmatrix}, \lambda \in \mathbb{R}$

Vector equation of plane $\pi$: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 10$

For the line to be parallel to the plane, $\mathbf{d}_\ell \cdot \mathbf{n}_\pi = 0$

$\begin{pmatrix} 2 \\ 1 \\ -\frac{b}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 0$

$6-1-2b=0$

$b = \frac{5}{2}$

For the line to not intersect the plane, the point must not lie on the plane,

i.e. $\begin{pmatrix} a \\ 2 \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \neq 10$

$3a-2+2 \neq 10$

| | |
|---|---|
| | $a \neq \frac{10}{3}$<br>The line and plane do not intersect when $a \neq \frac{10}{3}$ and $b = \frac{5}{2}$. |
| **(b)** | Equation of line: $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -\frac{3}{2} \end{pmatrix}$, $\lambda \in \mathbb{R}$<br>Let the position vector of the point of intersection, $P$ be<br>$\overrightarrow{OP} = \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -\frac{3}{2} \end{pmatrix} = \begin{pmatrix} 1+2\lambda \\ 2+\lambda \\ \frac{1}{2}-\frac{3}{2}\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$<br><br>Substituting into equation of plane $\pi$:<br>$\begin{pmatrix} 1+2\lambda \\ 2+\lambda \\ \frac{1}{2}-\frac{3}{2}\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 10$<br>$3(1+2\lambda) - (2+\lambda) + 4\left(\frac{1}{2} - \frac{3}{2}\lambda\right) = 10$<br>$3 + 6\lambda - 2 - \lambda + 2 - 6\lambda = 10$<br>$\therefore \quad \lambda = -7$<br>$\overrightarrow{OP} = \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} - 7\begin{pmatrix} 2 \\ 1 \\ -\frac{3}{2} \end{pmatrix} = \begin{pmatrix} -13 \\ -5 \\ 11 \end{pmatrix}$<br>The point of intersection $P$ is (–13, –5, 11). |
| **(c) (i)** | **Method 1: Using distance formula**<br>Taking the point $\left(1, 2, \frac{1}{2}\right)$ from the line $\ell$:<br>$\text{Distance} = \dfrac{\left\lvert \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} \cdot \mathbf{n} - d \right\rvert}{\lvert \mathbf{n} \rvert}$<br>$= \dfrac{\left\lvert \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} - 10 \right\rvert}{\sqrt{3^2 + (-1)^2 + 4^2}}$<br>$= \dfrac{7}{\sqrt{26}}$ units |

**Method 2: Finding length of projection on normal vector**

Taking the point $A\left(1, 2, \frac{1}{2}\right)$ from the line $\ell$, and the point $B$ (2, 0, 1) from the plane $\pi$:

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ \frac{1}{2} \end{pmatrix}$$

$$\text{Required distance} = \frac{\left| \begin{pmatrix} 1 \\ -2 \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \right|}{\left| \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \right|}$$

$$= \frac{|3+2+2|}{\sqrt{3^2+(-1)^2+4^2}}$$

$$= \frac{7}{\sqrt{26}} \text{ units}$$

**Method 3: Using distance between 2 parallel planes**

Consider a parallel plane containing the line $\ell$: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 3$

$$\text{Distance} = \left| \frac{10-3}{\sqrt{3^2+(-1)^2+4^2}} \right|$$

$$= \frac{7}{\sqrt{26}} \text{ units}$$

**(c) (ii)**

Equation of plane $\pi$: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 10 \Rightarrow \mathbf{r} \cdot \frac{\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}}{\sqrt{26}} = \frac{10}{\sqrt{26}}$

Equation of plane parallel to $\pi$ containing the line $\ell$: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = 3$

| | |
|---|---|
| | $\Rightarrow \mathbf{r} \cdot \dfrac{\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}}{\sqrt{26}} = \dfrac{3}{\sqrt{26}}$<br>Since they are of the same sign, plane $\pi$ and line $\ell$ are on the same side of the origin, therefore both the origin and the line $\ell$ lie on the same side of the plane. |
| | Another plane $\pi'$: $3x - y + 4z = -4$ |

**Source of Question: HCI CT2 9758/2023/01/Q6**

**8** The line $l$ has vector equation

$$\mathbf{r} = \begin{pmatrix} b \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ a \\ -5 \end{pmatrix},$$

where $\lambda$ is a real parameter and $a, b$ are real constants.

The equation of plane $\pi_1$ is given as $3x - y + 2z = 7$.

**(a)** Given that the distance between $l$ and $\pi_1$ is non-zero, find the value of $a$, and the set of values of $b$. [4]

For the rest of the question, let $a$ be the value found in part **(a)**, and $b = 2$.

**(b)** Find the vector equation of the line $l'$, where $l'$ is the reflection of $l$ in $\pi_1$. [4]

The equation of plane $\pi_2$ is given as $-3x + y - 2z = 2$.

**(c)** Given that the ratio of the distance between $\pi_1$ and $\pi_2$ to the distance between $\pi_2$ and plane $\pi_3$ is $2:3$, find the possible cartesian equations of $\pi_3$. [4]

**Solution:**

| | |
|---|---|
| **(a)** | Distance between $l$ and $\pi_1$ is non-zero $\Rightarrow$ $l$ and $\pi_1$ are parallel to each other, and $l$ does not lie on $\pi_1$.<br>$\pi_1 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$<br>$l$ and $\pi_1$ are parallel to each other $\Rightarrow$ $l$ is perpendicular to the normal of $\pi_1$<br><br>$\begin{pmatrix} 2 \\ a \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 0$<br>$6 - a - 10 = 0$<br>$a = -4$<br>$l$ does not lie on $\pi_1$ $\Rightarrow$ the point $(b, -1, 1)$ on $l$ does not lie on $\pi_1$ |

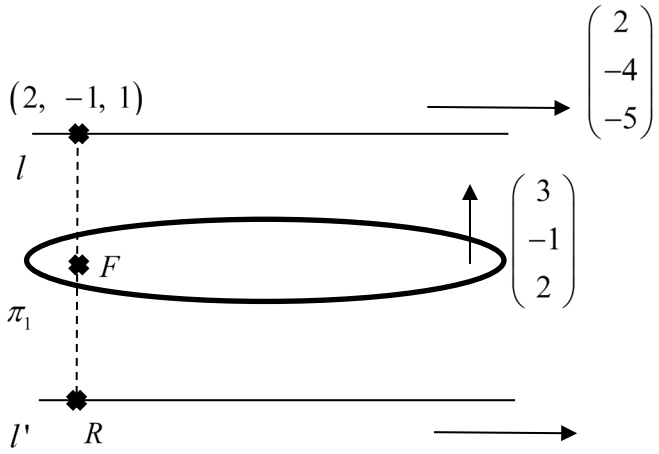
$\begin{pmatrix} b \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \neq 7$

$3b + 1 + 2 \neq 7$

$b \neq \frac{4}{3}$

The set of values of $b$ is $\left\{ b \in \mathbb{R} \middle| b \neq \frac{4}{3} \right\}$

**(b)** $a = -4$ implies that $l$ and $\pi_1$ are parallel to each other




The direction vector of $l'$ is parallel to $l$, i.e. a direction vector of $l'$ is $\begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$.

Let $F$ be the foot of perpendicular of point $(2, -1, 1)$ to $\pi_1$

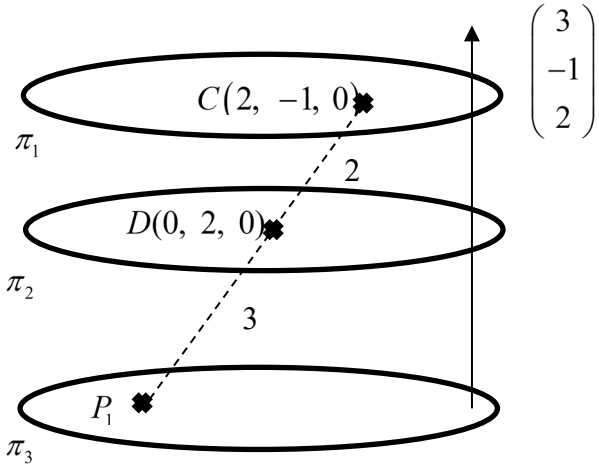
$\overrightarrow{OF} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ for some value of $\mu$

Since $F$ is on $\pi_1$, $\overrightarrow{OF} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7 \Rightarrow \begin{pmatrix} 2+3\mu \\ -1-\mu \\ 1+2\mu \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$

$\Rightarrow 6 + 9\mu + 1 + \mu + 2 + 4\mu = 7$

$\Rightarrow 14\mu = -2$

$\Rightarrow \mu = -\frac{1}{7}$

| | |
|---|---|
| | Therefore $\overrightarrow{OF} = \begin{pmatrix} 2+3\left(-\frac{1}{7}\right) \\ -1-\left(-\frac{1}{7}\right) \\ 1+2\left(-\frac{1}{7}\right) \end{pmatrix} = \begin{pmatrix} \frac{11}{7} \\ -\frac{6}{7} \\ \frac{5}{7} \end{pmatrix}$<br>Let $R$ be the reflection of $(2,\ -1,\ 1)$ about $\pi_1$<br>Using ratio theorem,<br>$\overrightarrow{OF} = \dfrac{\overrightarrow{OR} + \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}}{2} \Rightarrow \overrightarrow{OR} = 2\begin{pmatrix} \frac{11}{7} \\ -\frac{6}{7} \\ \frac{5}{7} \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{8}{7} \\ -\frac{5}{7} \\ \frac{3}{7} \end{pmatrix}$<br>Equation of line $l'$: $\mathbf{r} = \begin{pmatrix} \frac{8}{7} \\ -\frac{5}{7} \\ \frac{3}{7} \end{pmatrix} + \gamma \begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$, $\gamma \in \mathbb{R}$ |
| **(c)** | Since there is a non-zero distance between $\pi_2$ and plane $\pi_3$, $\pi_3$ is parallel to $\pi_2$ (and $\pi_1$).<br>Hence a normal to $\pi_3$ is $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$<br>$C(2,\ -1,\ 0)$ is on $\pi_1$,<br>and $D(0,\ 2,\ 0)$ is on $\pi_2$<br>  <br><br><u>Case 1: $\pi_2$ is between $\pi_1$ and $\pi_3$</u><br>Let $P_1$ be a point on $\pi_3$ such that<br>$CD : DP_1 = 2 : 3$<br>Using ratio theorem,<br>$\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \dfrac{2\overrightarrow{OP_1} + 3\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}}{5}$<br>$\Rightarrow 2\overrightarrow{OP_1} = 5\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} - 3\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \Rightarrow \overrightarrow{OP_1} = \begin{pmatrix} -3 \\ \frac{13}{2} \\ 0 \end{pmatrix}$<br>Equation of $\pi_3$: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ \frac{13}{2} \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = -9 - \dfrac{13}{2} + 0 = -\dfrac{31}{2}$ |

Cartesian equation of $\pi_3$: $3x-y+2z=-\dfrac{31}{2}$

Case 2: $\pi_1$ is between $\pi_2$ and $\pi_3$

Let $P_2$ be a point on $\pi_3$ such that

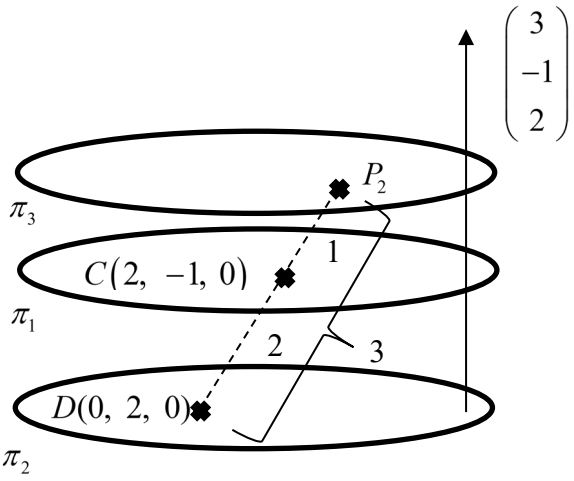
$CD:CP_2=2:1$

Using ratio theorem,

$$\begin{pmatrix}2\\-1\\0\end{pmatrix}=\frac{2\overrightarrow{OP_2}+\begin{pmatrix}0\\2\\0\end{pmatrix}}{3}$$

$$\Rightarrow 2\overrightarrow{OP_2}=3\begin{pmatrix}2\\-1\\0\end{pmatrix}-\begin{pmatrix}0\\2\\0\end{pmatrix}$$

$$\Rightarrow \overrightarrow{OP_2}=\begin{pmatrix}3\\-\frac{5}{2}\\0\end{pmatrix}$$




Equation of $\pi_3$: $\mathbf{r}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=\begin{pmatrix}3\\-\frac{5}{2}\\0\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=9+\dfrac{5}{2}+0=\dfrac{23}{2}$

Cartesian equation of $\pi_3$: $3x-y+2z=\dfrac{23}{2}$